

Haydeé Contreras Peruyero

Postdoctoral Research at
Centro de Ciencias
Matemáticas UNAM



18 Aug 1989



CCM UNAM



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<https://haydeeperuyero.github.io/>

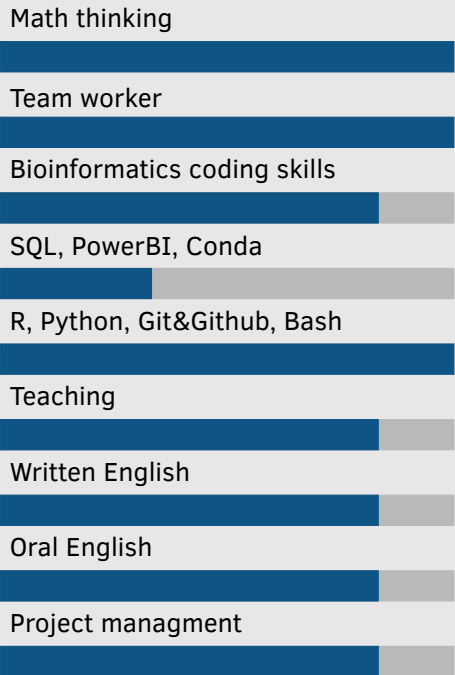


haydeeperuyero@matmor.unam.mx

About me

My research interests lie at the intersection of geometry, topology, and bioinformatics. My work has focused on geometric group theory and the application of Topological Data Analysis (TDA) to problems in genomics, metagenomics, and pangenomics. I also use machine learning and statistics to explore structures in biological data. I have a strong passion for teaching, programming, and learning languages. I was awarded the SNII Level 1 distinction in August 2025.

Skills



(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Interests

Topological ang Geometric Data Analysis | Bioinformatics | Geometric GroupTheory | Global Riemannian Geometry | Quantum Geometry and Topology

Education

2016-2021	Ph. D. in Science in Mathematics	Instituto de Matemáticas, UNAM
	Thesis: <i>Quasi-Isometric Rigidity in Higher Graph Manifolds</i>	
2020-2021	Specialty in Applied Statistics	Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas
2014-2016	M.Sc. in Mathematics	Instituto de Matemáticas, UNAM
	Thesis: On the Geometry and Topology of Graph Manifolds	
2007-2012	B.Sc. in Mathematics	Universidad Veracruzana
	Thesis: Triangulations of Hyperbolic 3-Manifolds Fibered over the Circle with Torus with a Hole as Fiber	

Professional Experience

2025-	Postdoctoral Research	CONAHCYTH, CCM UNAM
	<i>Topological Data Analysis Applied to Genomic Data and Cryptography with Geometric Group Theory</i>	
2023-2025	Postdoctoral research DGAPA	CCM UNAM
	Applications of Topological Data Analysis and Quasi-Isometry Invariants	
2022-2023	Postdoctoral research CONACyT Ciencia de Frontera	CONACyT, CCM UNAM
	Invariant Properties of Higher Graph Manifolds	
2021-2022	Head of the Department of Basic Sciences	ITSMT

Publications

2025	Topological data analysis captures Horizontal Gene Transfer in Antimicrobial Resistance gene families among clinically relevant bacteria. Guerrero-Flores et al. Front. Microbiol. 16:1461293. https://doi.org/10.3389/fmicb.2025.1461293
2025	Asymptotic dimension and geometric decompositions in dimensions 3 and 4, H. Contreras Peruyero and P. Suárez-Serrato. Journal of the Australian Mathematical Society. Published online 2025:1-26. https://doi.org/10.1017/S1446788725000072
2025	Linear bounds for the lengths of geodesics on manifolds with curvature bounded below. Accepted April 2025. https://arxiv.org/abs/2410.10975
2024	Multidisciplinary Workshop on Pangenomics and Topological Data Analysis, H. Contreras-Peruyero et al. Bioinformatics Advances, 2024;, vbae139 https://doi.org/10.1093/bioadv/vbae139
2024	CAMDA 2023: finding patterns in urban microbiomes. H. Contreras-Peruyero et al. Front. Genet. (2024) 15:1449461. https://doi.org/10.3389/fgene.2024.1449461
2024	Physicochemical Properties of Two Mexican Stingless Bee Honeys to Strengthen Their Biocultural Value. A. Xolalpa-Aroche et al. (2024). Revista Brasileira de Farmacognosia, 1-11. https://doi.org/10.1007/s43450-024-00566-z
2024	Actinobacteria bioprospecting from ore-forming environments. C. Aguilar et al. Microbial Genomics, 10(5), 001253. ISSN 2057-5858, https://doi.org/10.1099/mgen.0.001253
2022	Collapsing and group growth as obstructions to Einstein metrics on some smooth 4-manifolds. H. Contreras Peruyero and P. Suárez-Serrato, New York Journal of Mathematics 28 (2022), 659–671. http://nyjm.albany.edu/j/2022/28-27.html

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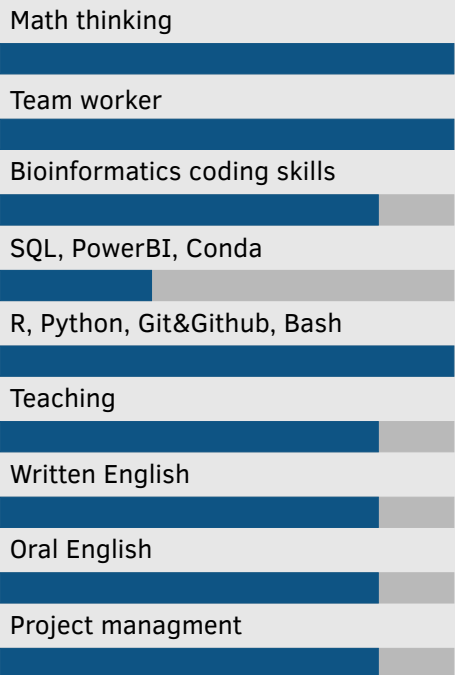
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Accepted papers

- 2025 Fractal dimensions of complex networks: advocating for a topological approach. Andreeva, R. et al. To appear at the AWM-IMA Springer volume “Research in Computational Topology”. <https://arxiv.org/abs/2506.15236>. Accepted May 2025.
- 2025 MicroAgroBiome: A toolkit for exploring specialized metabolism and ecological interactions in rhizosphere microbiomes of cultivated crops. Aguilar, C. et al. To appear at Nucleic Acids Research. Accepted September 2025.

Preprints

- 2025 Uncharacterized *Acetilactobacillus*-like genera represent the most abundant microorganisms in honeys produced by native stingless bee. Submitted June 2025. <https://doi.org/10.1101/2025.06.25.661387>
- 2024 Lengths of the Orthogonal Geodesic Chords on Riemannian Manifolds. In preparation. <https://arxiv.org/abs/2509.19620>
- 2022 On the quasi-isometric rigidity of chambers and walls in cusp-decomposable manifolds. Submitted 2022. <https://arxiv.org/abs/2010.03736>

Teaching Experience

- 2025 Undergraduate course “Multivariate Statistics” ENES Morelia UNAM
<https://haydeeperuyero.github.io/MultivariateStatisticalAnalysis/>
- 2024 Graduate course “Advanced Statistics Course: Machine Learning and Deep Geometric Learning” CCM UNAM
<https://github.com/CursosPosgradoCCM/ML-GDL>
- 2024 Graduate course “Scientific Computing: Scientific computing for data analysis” CCM UNAM
https://haydeeperuyero.github.io/Computo_Cientifico/
- 2023 Graduate course “Topology advanced course: Applications of Topological Data Analysis” CCM UNAM
https://github.com/HaydeePeruyero/CursoA_Aplicaciones_TDA
- 2023 Graduate course “Statistics seminar: Data Analysis with R” CCM UNAM
https://haydeeperuyero.github.io/Seminario_Estadistica/

Thesis Supervision

- 2024-2025 Master’s Thesis Advisor Johanna Atenea Carreón Baltazar CCM UNAM
Thesis title “Analysis of the Distribution of Duplicated Genes in Cyanobacteria”, Tesis Approved on October 3, 2025
- 2024- Master’s Thesis Co-advisor Luis Raul Figueroa Martinez CCM UNAM
Thesis title “Topological Data Analysis for Automated Classification of Multicellular Patterns”
- 2025- Undergraduate’s Thesis Advisor Alejandro Quintero Díaz FC. UNAM
Thesis title: “Aplicaciones de la homología persistente en el estudio de pangenomas y resistomasbacterianos”

Thesis Reviewer

- 2025 Master’s thesis reviewer of Jaime Salvador López Viveros CCM UNAM
Thesis title: “Identificación de recombinación en virus utilizando análisis topológicos de datos”
- 2025 Master’s thesis reviewer of Mariel Guadalupe Gutierrez Chaveste ccm UNAM
Thesis title: “Presentación de ADN mediante el juego de caos clasificación de imágenes y desarrollo de App de divulgación”

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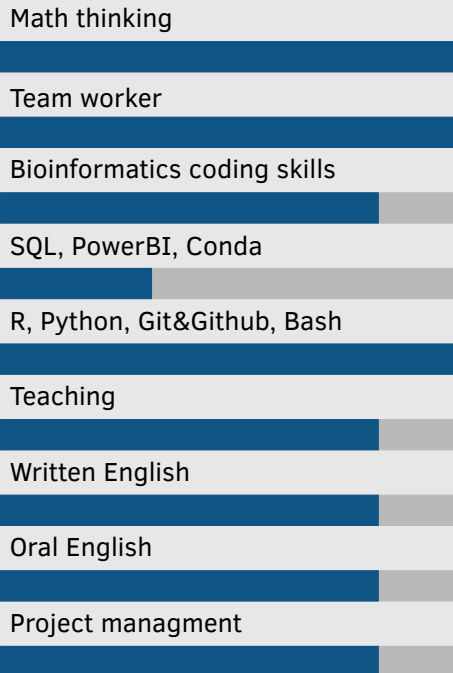


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Commities

2024-2025	Statistics General Knowledge Basic Exam Semester 2025-2, Semester 2025-1	CCM UNAM
2023-2024	Algebraic Topology General Knowledge Basic Exam Semester 2023-2, 2024-1, 2024-2	CCM UNAM

Selected talks and posters

2025	XVI Taller de Álgebra y Topología y 3° Jornada Estudiantil de Ciencias Físico-Matemáticas De datos biológicos a invariantes topológicos	Cuernavaca, Mexico
2025	Coloquio IMATE CU UNAM Explorando Datos Biológicos con Homología Persistente	CDMX, Mexico
2025	1st Brazil-Mexico Joint Mathematical Meeting Quasi-isometric rigidity on 4-manifolds	Fortaleza, Brazil
2025	4° Congreso Latinoamericano de Ecología Microbiana, ISME Lat 2025 Merida, Mexico Exploring honey microbiomes: a metagenomic dashboard for taxonomy, biosynthesis, and antibiotic resistance	
2025	Taller Mexicano de Topología Más allá de las formas: Cómo la topología revela patrones en datos biológicos	CDMX, Mexico
2025	CMO Workshop: Mixtures of probability and Geometry Detecting Evolutionary Patterns and Cellular Organization through Persistent Homology	Oaxaca, Mexico
2025	La Matemática Atenagorica Diversidad microbiana en mieles de abejas nativas sin aguijón de la Península de Yucatán	Michoacan, Mexico
2024	1st Asia&Pacific Bioinformatics Joint Conference Poster "Microbial Diversity and Characterization of Stingless Bee Honey from the Yucatán Peninsula"	Okinawa, Japan
2024	Escuela de Matemáticas de América Latina y el Caribe Cuernavaca 2024 Talk "Analizando datos con topología y geometría"	Morelos, Mexico
2024	2024 Joint Mathematics Meetings Talk "Asymptotic dimension and geometric decompositions in dimension 4"	San Francisco, USA
2023	XXIV Escuela de Otoño (EOBM) y el XVII Encuentro Nacional (ENBM), of Bio-mathematics Minicourse "Aplicaciones de análisis topológico de datos a Pangenómica"	Veracruz, Mexico
2023	Iberoamerican and Pan Pacific International Conference on Topology and its Applications "Invariants under quasi-isometries"	Puebla, Mexico
2023	31st Conference on Intelligent Systems for Molecular Biology and 22th European Conference On Computational Biology as part of the Annual International Conference on Critical Assessment of Massive Data Analysis Poster "Topology, regression and metapangenome enrich conventional microbiome analyses"	Lyon, France
2023	2023 Joint Mathematics Meetings Talk "Collapsing and group growth as an obstruction to Einstein metrics on some smooth 4-manifolds"	Boston, USA
2022	8th Mexican Workshop on Applied Geometry and Topology Poster "Topological and Geometric explorations of Autism fMRI connectomes"	Virtual

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Skills

Math thinking

Team worker

Bioinformatics coding skills

SQL, PowerBI, Conda

R, Python, Git&Github, Bash

Teaching

Written English

Oral English

Project managment

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Recent Awards

2025	Investigador Nacional Nivel I, Sistema Nacional de Investigadoras e Investigadores. August 2025
2023	1 Price. Critical Assessment of Massive Data Analysis Microbiome Challenge. CAMDA 2023
2022	Mentee in the Mentoring in Action - Women and Girls in STEM Online programme
2022	2nd place at the ISCB-La SOLBIO BIONETMX WIKIPEDIA COMPETITION, Article: "Matrices de distancias"

Science communication

2025	Tianguis de las Ciencias de la Universidad Michoacana de San Nicolas de Hidalgo Workshop "Papiroflexia Modular"	Michoacan, Mexico
2024 - 2025	Feria de las Matemáticas Workshop "Papiroflexia Modular"	Michoacan, Mexico
2024	Futuras STEM Michoacán Workshop "Matemagia"	Michoacan, Mexico
2024	Feria Infantil y Juvenil de Ciencias y Artes Workshop "Estereogramas"	Michoacan, Mexico
2024	Feria de las Matemáticas Talk "¿Qué hacemos cuentas? Son puros cuentos"	Michoacan, Mexico
2023	11 Fiesta de las Ciencias y Humanidades Workshop "Lengua de Señas Mexicana (LSM) y Matemáticas"	Michoacan, Mexico
2023	11 Fiesta de las Ciencias y Humanidades Workshop "Biomatemáticas"	Michoacan, Mexico

Synergistic Activities

2024 - 2025	Encuentro de Mujeres en Topología, Geometrpia y sus Aplicaciones Michoacan, Mexico 1st and 2nd, Co-organizer, Webpage: https://matmor.unam.mx/encuentro-mujeres-topogeo-2025/	
2023 - 2025	Escuela de Teoría Geométrica de Grupos II, III, IV, Co-organizer, webpage https://matmor.unam.mx/etgg-2025/	Michoacan, Mexico
2023 - 2025	Hackaton Reto Critical Assessment of Massive Data Co-organizer 2023, 2024 and 2025	Michoacan, Mexico
2022-	Long-distance Seminar on Geometric Group Theory in Mexico Co-organizer	Virtual
2022-	R Ladies Morelia Co-organizer, webpage https://www.meetup.com/es-ES/rladies-morelia/?eventOrigin=home_groups_you_organize	Hybrid, Michoacan, Mexico